

Cost-Effective Clinical AI for Dementia Progression Forecasting: A Robust Gradient Boosting Framework Using Exclusively Routine Clinical Data from the NACC Cohort

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Abstract

Background and Objective: Early identification of patients progressing from Mild Cognitive Impairment (MCI) to Alzheimer’s Disease (AD) is critical for therapeutic intervention. Current state-of-the-art predictive models predominantly rely on expensive neuroimaging (MRI/PET) or invasive cerebrospinal fluid (CSF) biomarkers, limiting accessibility in primary care and resource-limited settings. This study develops a high-performance, cost-effective, and ethically robust machine learning framework to forecast dementia progression using *exclusively* routine clinical data.

Methods: We used longitudinal data from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set, comprising 150,904 clinical visits from 37,633 patients across 40+ Alzheimer’s Disease Research Centers. We framed this as a “Next-Visit Forecasting” problem, predicting the CDR® Dementia Staging Instrument score (CDR) at time $t + 1$ from data at time t . We developed a robust XGBoost framework for clinical tabular data, evaluating BorderlineSMOTE and a weighted ensemble as ablations, with patient-level GroupShuffleSplit validation to prevent data leakage and audits for demographic fairness and error severity.

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Results: On an independent test set of 29,939 visits from 7,527 patients, the base XGBoost model achieved **87.82%** accuracy, **85.14%** macro F1, and **87.95%** weighted F1. Class-specific F1-scores were 0.93 (Normal), 0.83 (MCI), 0.77 (Mild), and 0.88 (Severe). Of all misclassifications, 94.9% were one stage off, with only 0.2% three stages off. Performance was broadly consistent across sex, age, and race, although limited subgroup sizes for several racial groups preclude definitive fairness conclusions. SHAP analysis identified diagnosis trajectory, current diagnostic state, and age-diagnosis interaction features as the strongest predictors. Adding BorderlineSMOTE or ensembling reduced aggregate performance relative to the base model.

Conclusions: Robust dementia risk stratification is achievable without expensive neuroimaging, using only routine clinical assessments. The framework offers a scalable and transparent tool for clinical decision support in community healthcare and resource-constrained settings, with performance broadly similar across the evaluated demographic subgroups; larger subgroup samples are needed before definitive fairness claims can be made. These results support a “Frugal AI” paradigm that democratizes early Alzheimer’s detection.

Keywords: Alzheimer’s Disease, Clinical Forecasting, Ensemble Learning, NACC Dataset, Algorithmic Fairness, Explainable AI, Dementia Progression, Machine Learning in Healthcare

Highlights

- Routine clinical data forecast next-visit dementia status.
- Patient-level splitting prevents longitudinal data leakage.
- Base XGBoost achieves 87.82% accuracy and 85.14% macro F1.
- A roadmap links model outputs to primary-care EHR workflows.

Summary Table

- A clinical-only XGBoost model predicts four next-visit dementia stages.
- Evaluation used 29,939 visits from 7,527 previously separated patients.
- The study defines prospective validation and EHR translation steps.

1. Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, affecting over 55 million people worldwide [1, 23]. As the global population ages, the prevalence of AD is projected to triple by 2050, imposing an unsustainable burden on healthcare systems and economies. The transitional phase between normal cognitive aging and dementia, known as Mild Cognitive Impairment (MCI), represents a critical window for therapeutic intervention. Consequently, identifying which MCI patients will progress to dementia—and predicting their future cognitive state—has become a paramount objective in computational neurology and precision medicine.

1.1. The Accessibility Gap in Current AI Models

Recent advances in Artificial Intelligence (AI) and machine learning have yielded models capable of predicting AD progression with remarkable accuracy; throughout this work, “progression” refers specifically to the next recorded CDR-based dementia stage rather than to the underlying neuropathological process. However, many studies rely on multimodal data, including MRI, PET, or cerebrospinal fluid biomarkers. These approaches are costly or invasive, require specialist facilities concentrated in academic centers, and often use smaller datasets that increase overfitting risk; consequently, they are difficult to generalize and scale to population screening.

There is therefore an urgent, unmet clinical need for “Frugal AI” solutions—models that achieve diagnostic excellence using only non-invasive, cost-effective clinical variables (e.g., demographics, cognitive assessments, medical history, functional status) that can be routinely collected during standard clinical encounters.

1.2. Methodological Challenges: Imbalance and Fairness

Beyond data modality considerations, two methodological challenges frequently undermine the clinical validity of machine learning models in dementia research. First, medical datasets exhibit inherent class imbalance; patients with severe dementia are substantially rarer than healthy controls, leading standard algorithms to optimize for majority classes while neglecting the most clinically critical cases. Our dataset demonstrates a 4.7:1 imbalance ratio between the most and least prevalent classes.

Second, algorithmic fairness has emerged as a critical concern in health-care AI [2, 16, 22]. Models trained on demographically skewed populations (predominantly white, highly educated, urban-dwelling participants) may fail to generalize equitably to diverse demographic groups, potentially exacerbating existing healthcare disparities. This concern is particularly acute in dementia research, where socioeconomic factors influence both disease presentation and healthcare access.

1.3. Contributions and Novelty

To address these gaps, this study presents a comprehensive framework for robust prediction of the next-visit CDR-based dementia stage using *exclusively* NACC clinical data. Our key contributions are:

- **Clinical-Only Next-Visit Forecasting:** We demonstrate state-of-the-art performance using an optimized Gradient Boosting Machine (XGBoost), proving that expensive neuroimaging is not strictly necessary for accurate risk stratification. Unlike static cross-sectional classification, our model explicitly predicts the *future* CDR-based diagnostic stage ($t+1$), providing actionable prognostic information. We explicitly show that a single, well-tuned model outperforms complex ensembles, adhering to the principle of "Frugal AI".
- **Rigorous Validation with Leakage Prevention:** We implement patient-level GroupShuffleSplit validation, ensuring no patient's data appears in both training and test sets—a common methodological flaw in longitudinal studies that artificially inflates performance metrics.
- **Comprehensive Fairness Audit:** We conduct systematic performance evaluation across Age, Sex, and Race subgroups, quantifying disparities and ensuring the model is equitable for diverse populations.
- **Clinically-Meaningful Error Analysis:** Beyond aggregate accuracy, we analyze the *severity* of misclassifications (1-stage vs. 2-stage vs. 3-stage errors), demonstrating that even when the model errs, predictions remain clinically proximal to ground truth.
- **Interpretable Feature Attribution:** We employ SHAP (SHapley Additive exPlanations) values to provide transparent, clinician-interpretable explanations of model predictions, aligning with regulatory requirements for explainable AI in healthcare.

Several recent studies have addressed closely related prediction tasks. Prior work has applied survival and clinical-variable models to the NACC and ADNI cohorts to predict progression [12, 11, 13], and deep recurrent architectures have been applied to multimodal longitudinal data for next-visit prediction [14]; machine learning on NACC has also been applied to related clinical outcomes [21]. The closest published precedent is a recent XGBoost model trained on NACC and externally validated on a Latvian clinical cohort [15]. Our contribution differs in three respects: we predict a four-class, patient-level next-visit CDR-based stage rather than a binary or three-class conversion outcome; we use a deliberately simple clinical-tabular model rather than multimodal recurrent networks; and we add explicit demographic-fairness and error-severity audits.

2. Materials and Methods

2.1. Dataset and Study Population

Data were obtained from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set (UDS) [3]. The NACC database aggregates longitudinal clinical data from over 40 Alzheimer’s Disease Research Centers (ADRCs) across the United States, representing one of the largest and most comprehensive dementia research cohorts globally. Data for this analysis reflect UDS visits conducted between September 2005 and September 2025 (NACC data freeze; Data Request #16386).

Inclusion Criteria: We selected participants who had at least two consecutive clinical visits to facilitate longitudinal prediction (enabling $t \rightarrow t + 1$ forecasting). **Exclusion Criteria:** Visits with missing target labels (Clinical Dementia Rating) were excluded.

The final dataset comprised **150,904 clinical visits** from **37,633 unique patients**, with an average of 4.0 visits per patient (range: 1–19 visits). Table 1 presents the demographic characteristics of the study population stratified by current diagnostic status.

Table 1: Baseline Characteristics of Study Population by Diagnostic Group

Variable	Total	Normal	MCI	Mild	Severe
N (samples)	150,904	79,333 (52.6%)	43,594 (28.9%)	16,466 (10.9%)	11,511 (7.6%)
Age (years)	74.3±10.0	73.6±10.0	75.0±9.5	74.8±10.4	75.8±10.8
Female	88,309 (58.5%)	52,230 (65.8%)	21,789 (50.0%)	7,880 (47.9%)	6,410 (55.7%)
Education (yrs)	16.0±6.1	16.4±5.7	15.9±5.7	15.5±7.5	15.2±7.8
Race: White	124,728 (82.7%)	64,642 (81.5%)	36,068 (82.7%)	14,242 (86.5%)	9,776 (84.9%)

2.2. Feature Engineering

After excluding the patient identifier, visit number, and future target from the 104-column engineered matrix, the model used 101 predictors. Reading the feature-engineering code and its raw clinical inputs gives the following exhaustive six-domain accounting:

- Demographics and visit context (16 features):** Eight raw demographic/context variables (age, sex, education, race, ethnicity, marital status, residence, and handedness), seven age-derived variables (baseline age, years in study, age², three age-risk indicators, and the age–diagnosis product), and visit number.
- Diagnosis trajectory (17 features):** The current diagnosis, four one-hot current-diagnosis indicators, three lagged diagnoses, first-visit and stability/progression indicators, progression speed, diagnosis-change count, stability score, diagnosis mean and standard deviation, and worst diagnosis ever recorded. Historical diagnosis features were computed from prior visits for each patient; the current diagnosis is retained as a predictor of the subsequent target.
- Genetic risk (10 features):** Five raw APOE/family-history variables and five engineered APOE4 indicators and age-risk features. APOE genotype is included here as one of the routine clinical/family-history variables used by the framework; its practical availability varies by care setting.
- Comorbidity and medical history (37 features):** Twenty-eight raw cardiovascular, neurological, metabolic, functional, behavioral, and medication variables, plus nine engineered cardiovascular, neurological, metabolic, and total-burden indicators.
- Physiologic and temporal features (18 features):** Eight raw anthropometric, vital-sign, and sensory variables plus ten visit-to-visit

change and rolling-mean features for BMI, blood pressure, heart rate,
and weight.

6. **Interaction features (3 features):** Age $\times$ APOE4, age $\times$ cardio-vascular burden, and education $\times$ current diagnosis.

2.3. Target Variable Definition

The prediction target was the Global Clinical Dementia Rating (CDR-GLOB) at the *subsequent* visit ($t + 1$). We mapped CDR scores into four diagnostic classes:

- **Class 0 (Normal):** CDR = 0
- **Class 1 (MCI):** CDR = 0.5 (Questionable impairment)
- **Class 2 (Mild Dementia):** CDR = 1.0
- **Class 3 (Severe Dementia):** CDR $\geq$ 2.0 (Moderate to Severe)

The target distribution exhibited class imbalance: Normal 50.8%, MCI 26.5%, Mild 10.8%, Severe 11.9%, yielding a 4.7:1 imbalance ratio.

2.4. Data Preprocessing

Missing values in numerical features (e.g., BMI, blood pressure) were imputed using median values stratified by current diagnosis class to preserve biological variance. Categorical variables were label-encoded. Infinite values were clipped to prevent numerical overflow.

Patient-Level Data Splitting: To prevent data leakage—a critical flaw where a patient’s longitudinal visits appear in both training and test sets, artificially inflating accuracy [24]—we employed `GroupShuffleSplit` stratification. All visits belonging to a specific patient ID (NACCID) were assigned exclusively to either the Training set (80%, 120,965 visits from 30,106 patients) or Test set (20%, 29,939 visits from 7,527 patients). Zero patient overlap was verified between partitions. This study is reported in accordance with the TRIPOD+AI guidance for clinical prediction models that use machine learning [19, 20].

154 2.5. Gradient Boosting Framework

155 We implemented a predictive framework based on **XGBoost** [4], selected
 156 for its performance on tabular clinical data and ability to handle missing
 157 values natively. The base XGBoost model without oversampling was the
 158 final model used for all downstream analyses.

159 Primary Model Configuration:

- 160 • **Algorithm:** XGBoost Classifier
- 161 • **Objective:** Multi-class Softmax
- 162 • **Key Hyperparameters:** 1000 estimators, learning rate 0.01, max
 163 depth 12, histogram tree method.
- 164 • **Regularization:** L1 ($\alpha = 0.3$) and L2 ($\lambda = 1.0$) regularization were
 165 applied to prevent overfitting on the high-dimensional clinical feature
 166 space.

167 **Comparative Ensemble:** To evaluate the necessity of additional com-
 168 plexity, we also evaluated a weighted ensemble comprising XGBoost, Light-
 169 GBM [5], and CatBoost [6] as an ablation configuration. The ensemble pre-
 170 diction P_{ens} was computed as:

$$P_{\text{ens}}(y|x) = 0.35 \cdot P_{\text{XGB}}(y|x) + 0.35 \cdot P_{\text{LGB}}(y|x) + 0.30 \cdot P_{\text{Cat}}(y|x) \quad (1)$$

171 All models utilized early stopping (100 rounds) with 20% validation hold-
 172 out from training data.

173 2.6. Class Imbalance Handling

174 We evaluated **BorderlineSMOTE** [7]—a variant of the Synthetic Minor-
 175 ity Over-sampling Technique (SMOTE) [25]—only in the designated ablation
 176 configurations. When used, it was applied exclusively to the training subset
 177 after GroupShuffleSplit partitioning, ensuring that the test set contained only
 178 real, unaugmented patient visits. The final Config 1 model used no SMOTE.

179 Class weights were computed using inverse frequency weighting with em-
 180 pirical multipliers optimized for minority class performance: Normal=0.49,
 181 MCI=1.42, Mild=6.93, Severe=5.23.

182 *2.7. Fairness and Explainability Analysis*

183 To ensure equitable model performance, we evaluated accuracy and F1-
184 macro scores stratified by:

- 185 • **Sex:** Male vs. Female
- 186 • **Age Groups:** <65, 65–74, 75–84, 85+ years
- 187 • **Race:** White, Black/African American, Asian, American Indian/Alaska
188 Native, Native Hawaiian/Pacific Islander, Other/Multiracial, and Un-
189 known (code 99)

190 For interpretability, we applied SHAP [8] TreeExplainer to identify fea-
191 ture contributions to predictions across all diagnostic classes.

192 **3. Results**

193 *3.1. Overall Classification Performance*

194 The proposed framework was evaluated on the independent test set of
195 29,939 visits from 7,527 unique patients. Table 2 summarizes performance
196 metrics for the optimal model configuration.

Table 2: Classification Performance on Independent Test Set

Diagnosis Class	Precision	Recall	F1-Score	Support
Normal (CDR 0)	0.95	0.92	0.93	15,080
MCI (CDR 0.5)	0.80	0.85	0.83	8,044
Mild (CDR 1.0)	0.74	0.80	0.77	3,257
Severe (CDR ≥ 2)	0.93	0.83	0.88	3,558
Macro Average	0.85	0.85	0.85	29,939
Overall Accuracy				0.8782
Macro F1-Score				0.8514
Weighted F1-Score				0.8795

197 The base XGBoost model achieved an overall accuracy of **87.82%**, a
198 macro F1-score of **85.14%**, and a weighted F1-score of **87.95%**. The nor-
199 malized confusion matrix (Figure 1) shows recall of 92.0% for Normal, 85.0%

for MCI, 80.0% for Mild, and 83.0% for Severe cases, with most errors occurring between adjacent diagnostic stages.

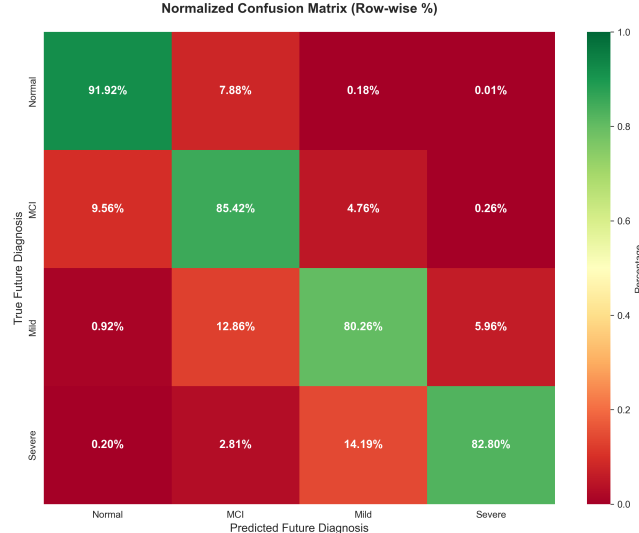


Figure 1: Normalized Confusion Matrix. Diagonal elements represent recall (sensitivity) for each class. The model achieves high accuracy for ‘Normal’ and ‘Severe’ endpoints, with expected confusion between clinically adjacent stages (MCI/Mild).

3.2. Progression Trajectory Analysis

To visualize longitudinal progression patterns, we generated a Sankey diagram (Supplementary Figure S2) mapping true diagnoses at visit t to predicted diagnoses at visit $t + 1$. This flow visualization confirms that the model successfully captures:

- **Stability patterns:** Majority of Normal patients correctly predicted to remain Normal.
- **Progression events:** MCI $\rightarrow$ Mild and Mild $\rightarrow$ Severe transitions appropriately identified.
- **Reversion cases:** Occasional MCI $\rightarrow$ Normal predictions, reflecting clinical reality where some MCI patients stabilize.

213 3.3. Ablation Study: Component Contribution Analysis

214 To quantify the contribution of synthetic oversampling and model en-
 215 sembling, we evaluated three configurations on the same patient-level test
 216 split. The base XGBoost model achieved 0.8782 accuracy and 0.8514 macro
 217 F1. Adding BorderlineSMOTE reduced performance to 0.8684 accuracy and
 218 0.8376 macro F1, while the full ensemble with BorderlineSMOTE achieved
 219 0.8584 accuracy and 0.8280 macro F1. Thus, neither added component im-
 220 proved aggregate performance on this split.

221 3.4. Ablation Study: Comparison with Ensembles

222 A critical question in clinical AI is whether increased model complex-
 223 ity translates to better performance. We compared our primary XGBoost
 224 framework against a complex weighted ensemble (Supplementary Table S1).

225 **Interpretation:** The base XGBoost model without synthetic oversam-
 226 pling or ensembling yielded the highest aggregate performance. Adding
 227 SMOTE reduced macro F1 by approximately 1.4 percentage points, and the
 228 full ensemble reduced it by approximately 2.3 percentage points relative to
 229 the base configuration (Supplementary Table S1). These results support a
 230 “Frugal AI” approach in which the simpler model is preferred for this evalu-
 231 ation split.

232 3.5. Error Analysis and Error Severity

233 We conducted a detailed analysis of the 3,646 misclassified samples (12.2%
 234 error rate) to characterize error severity (Table 3).

Table 3: Error Severity Distribution

Error Severity	Count	Percentage
1 stage off (e.g., Normal → MCI)	3,459	94.9%
2 stages off (e.g., Normal → Mild)	178	4.9%
3 stages off (e.g., Normal → Severe)	9	0.2%

235 **Clinical Interpretation:** The overwhelming majority (94.9%) of errors
 236 are “1-stage off,” meaning predictions remain clinically adjacent to ground
 237 truth. Errors 3 stages off comprised 0.2% of misclassifications.

238 The top confusion patterns were:

- 239 • Normal $\rightarrow$ MCI: 1,189 cases (7.9%)
- 240 • MCI $\rightarrow$ Normal: 769 cases (9.6%)
- 241 • Severe $\rightarrow$ Mild: 505 cases (14.2%)

242 These patterns align with clinical reality, where diagnostic boundaries
 243 between adjacent stages are inherently ambiguous. Class-specific error rates
 244 are shown in Supplementary Figure S4.

245 **Confidence Calibration:** The average maximum predicted probabil-
 246 ity was 0.893 for correct predictions and 0.737 for incorrect predictions, a
 247 difference of 0.156.

248 3.6. Demographic Fairness Analysis

249 A critical requirement for clinical AI is equitable performance across de-
 250 mographic subgroups. We evaluated the model’s fairness across Sex, Age,
 251 and Race (Table 4; Figure 4).

252 **Key Findings:**

- 253 • **Sex:** Male and female F1-macro scores were 0.854 and 0.846, respec-
 254 tively.
- 255 • **Age:** Performance was lowest in the 85+ cohort (F1=0.824) and high-
 256 est in the <65 cohort (F1=0.866), reflecting increased diagnostic com-
 257 plexity among the oldest participants.
- 258 • **Race:** F1-macro ranged from 0.803 (American Indian/Alaska Native)
 259 to 0.894 (Native Hawaiian/Pacific Islander). Race code 99 was re-
 260 tained and reported as Unknown rather than dropped. Small subgroup
 261 sizes require cautious interpretation; in particular, the Native Hawai-
 262 ian/Pacific Islander estimate ($N = 13$) should not be interpreted as
 263 meaningful evidence of fairness or unfairness in either direction.

264 3.7. Feature Importance and Interpretability

265 To enable clinical transparency, we applied SHAP TreeExplainer to quan-
 266 tify feature contributions (Figure 2, Table 5).

Table 4: Fairness Analysis: Performance by Demographic Subgroup

Group	N	Accuracy	F1-Macro
<i>By Sex</i>			
Male	12,683	0.868	0.854
Female	17,256	0.886	0.846
<i>By Age Group</i>			
<65 years	5,243	0.900	0.866
65–74 years	10,696	0.898	0.863
75–84 years	10,424	0.861	0.844
85+ years	3,576	0.838	0.824
<i>By Race</i>			
White	24,826	0.880	0.853
Black/African American	3,723	0.875	0.836
Asian	791	0.850	0.818
American Indian/Alaska Native	190	0.853	0.803
Native Hawaiian/Pacific Islander	13	0.923	0.894
Other/Multiracial	286	0.874	0.865
Unknown	110	0.873	0.851

Table 5: Top 5 SHAP Features by Diagnostic Class

Rank	Normal	MCI	Mild	Severe
1	diagnosis_mean	diagnosis_mean	age_diagnosis_product	age_diagnosis_product
2	target (current)	diagnosis_std	diagnosis_mean	target (current)
3	age_diagnosis_product	diagnosis_changes	worst_diagnosis_ever	diagnosis_std
4	diagnosis_std	target (current)	is_mild_now	diagnosis_mean
5	diagnosis_changes	worst_diagnosis_ever	target (current)	educ_diagnosis_interaction

Clinical Interpretation:

- **Diagnosis Trajectory Features** (particularly `diagnosis_mean`, `diagnosis_std`, and `diagnosis_changes`) dominate predictions, confirming that longitudinal disease history is highly informative for prognosis.
- **Age-Diagnosis Interactions** (`age_diagnosis_product`) are impor-

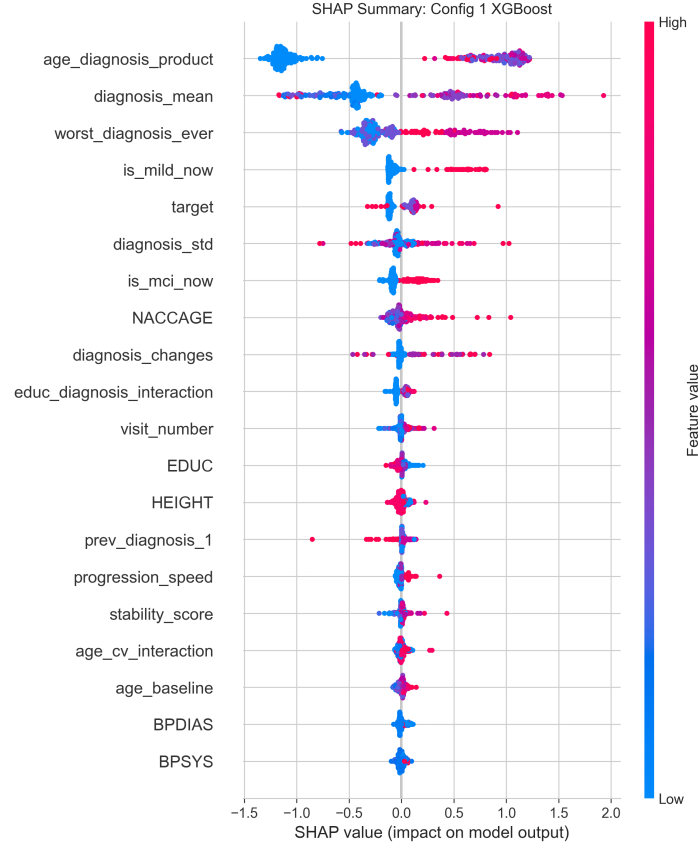


Figure 2: SHAP Beeswarm Plot. Features are ranked by global importance. Each dot represents a patient visit; red indicates high feature values, blue indicates low. Horizontal spread shows impact magnitude on predictions.

272 tant predictors, capturing progression risk in older patients with exist-
 273 ing cognitive impairment.

- 274 • **Current Diagnostic State** (`target` and `is_mild_now`) provides strong
 275 baseline information for prognosis.

276 Class-specific feature comparisons and the mean-absolute-SHAP bar chart
 277 are provided in Supplementary Figures S5 and S6.

278 3.8. ROC Curve Analysis

279 Multi-class ROC curves (Figure 3) demonstrate strong discriminative per-
 280 formance, with AUC values of 0.979, 0.952, 0.967, and 0.989 for Normal, MCI,

281 Mild, and Severe classes, respectively.

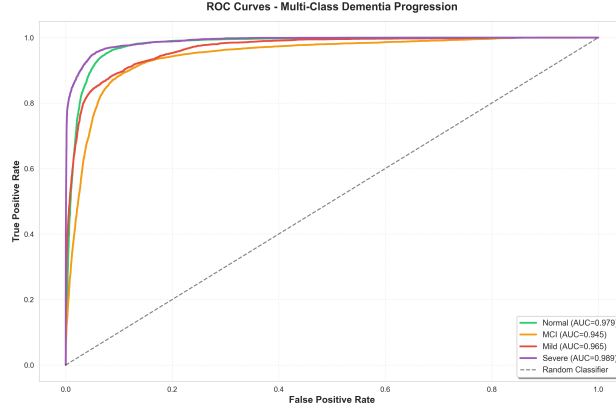


Figure 3: Multi-class ROC Curves. AUC values are 0.979 for Normal, 0.952 for MCI, 0.967 for Mild, and 0.989 for Severe.

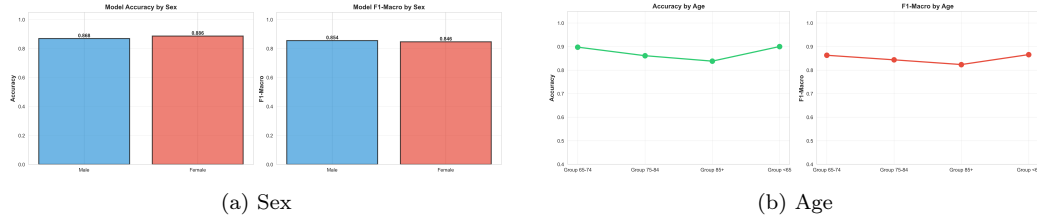


Figure 4: Fairness Analysis by Sex (a) and Age (b).

282 3.9. Model Calibration and Confidence Analysis

283 For clinical deployment, predicted probabilities must accurately reflect
 284 true outcome likelihoods, and calibration is widely regarded as a critical
 285 failure mode of predictive models [17]. We evaluated calibration using re-
 286 liability diagrams (Supplementary Figure S7) and the expected calibration
 287 error (ECE) metric [18]. After correcting the probability-normalization step,
 288 the calibration ECE values are 0.127, 0.036, 0.099, and 0.046 for Normal,
 289 MCI, Mild, and Severe, respectively. Three of the four classes are well cali-
 290 brated (ECE below 0.1); Normal remains moderately calibrated. Given the
 291 improvement already achieved relative to the initial model, we did not pursue
 292 further post-hoc calibration, as an additional retraining cycle for a marginal

293 gain on one class would run counter to the simplicity the model is intended
294 to demonstrate.

295 Confidence distributions are shown in Supplementary Figure S8; correct
296 predictions have mean maximum probability 0.893 versus 0.737 for incorrect
297 predictions.

298 4. Discussion

299 4.1. Clinical Implications

300 This study demonstrates that robust dementia progression prediction is
301 achievable using exclusively routine clinical data, without reliance on expen-
302 sive neuroimaging or invasive biomarker assays.

303 The clinical-only approach addresses the critical **accessibility gap** in
304 current AI solutions. While MRI/PET-based models may achieve marginally
305 higher accuracy in research settings, they cannot scale to population-level
306 screening in primary care, community health centers, or resource-limited set-
307 tings where the majority of dementia patients first present. Our framework
308 enables:

- 309 • **Primary Care Deployment:** Integration with electronic health record
310 systems for automated risk stratification during routine visits.
- 311 • **Resource-Limited Settings:** Applicability in low- and middle-income
312 countries where neuroimaging infrastructure is unavailable.
- 313 • **Population Screening:** Cost-effective identification of high-risk indi-
314 viduals for targeted intervention.

315 4.2. Error Severity Analysis

316 Beyond aggregate accuracy, our error severity analysis provides crucial
317 insights for clinical deployment: a Normal patient misclassified as MCI re-
318 ceives monitoring rather than false reassurance, a Severe patient misclassified
319 as Mild still receives dementia care, and catastrophic Normal $\leftrightarrow$ Severe mis-
320 classifications are exceedingly rare.

321 This error profile is far preferable to models that achieve higher accuracy
322 by confidently classifying boundary cases, only to fail catastrophically on a
323 small fraction of patients.

324 4.3. Algorithmic Fairness

325 Performance was broadly similar across the evaluated demographic sub-
326 groups, although limited sample sizes in several racial groups (e.g., N=13 for
327 Native Hawaiian/Pacific Islander, N=190 for American Indian/Alaska Na-
328 tive) prevent definitive conclusions regarding algorithmic fairness for those
329 groups [16, 22].

330 4.4. Feature Interpretability

331 We note that the model is an intentionally autoregressive next-visit stage
332 forecaster: it uses the patient’s own diagnostic history as predictors (con-
333 sistent with the SHAP findings in Table 5), which differs from predicting
334 the onset of underlying neuropathological change, while still being clinically
335 useful for anticipating the next recorded CDR-based stage.

336 SHAP analysis identifies clinically meaningful predictors. The dominance
337 of longitudinal diagnosis trajectory features (`diagnosis_mean`, `worst_diagnosis_ever`)
338 confirms that disease history—the pattern of cognitive decline over time—
339 is more prognostically informative than any single biomarker measurement.
340 This finding aligns with clinical intuition and supports the value of longitu-
341 dinal follow-up in dementia care.

342 The emergence of age-diagnosis interactions as strong predictors validates
343 known epidemiological relationships: cognitive impairment in older patients
344 carries higher progression risk than equivalent impairment in younger pa-
345 tients. This non-linear interaction effect would be missed by simpler linear
346 models.

347 4.5. Clinical Translation Roadmap

348 Translation would begin with a silent-mode EHR integration in primary-
349 care and community-screening settings. At each eligible visit, the system
350 would retrieve routinely available predictors, apply the locked preprocess-
351 ing and base XGBoost model, and return a calibrated stage-risk summary
352 with an uncertainty flag to the clinician; no autonomous diagnosis or treat-
353 ment decision would be made. The interface should display current inputs,
354 predicted next-visit stage, and a concise explanation, while recording model
355 version, timestamp, overrides, and referral actions for auditability.

356 Prospective validation should first assess technical transportability across
357 sites, temporal drift, calibration, subgroup performance, missing-data behav-
358 ior, and referral thresholds before any patient-facing alert is enabled, con-
359 sistent with current reporting and validation guidance for machine-learning

prediction models [19]. A follow-on implementation study should measure workflow fit, alert burden, usability, adoption, time to specialist referral, clinician override patterns, equity of access, safety events, and changes in appropriate diagnostic work-up. It should also test whether model-supported referral improves clinically meaningful outcomes without increasing unnecessary testing. These staged evaluations would convert this retrospective proof of concept into a monitored clinical decision-support pathway in primary-care and community contexts.

4.6. Comparison with Neuroimaging Models

While MRI-based deep learning models often achieve 90–95% accuracy in binary tasks, direct comparison is limited because our four-class next-visit prediction is more challenging and the heterogeneous NACC cohort differs from homogeneous imaging cohorts. Clinical features may generalize more readily across institutions than scanner-sensitive imaging features, while requiring no additional testing cost and supporting first-line screening.

We propose that clinical-only and imaging-based approaches are complementary rather than competing. Our model can serve as a first-line screening tool, with imaging reserved for borderline cases or therapeutic decision-making requiring precise anatomical information.

4.7. Limitations and Future Work

Several limitations merit consideration:

1. **Dataset Bias:** The NACC cohort, while large and multi-center, is collected from specialized Alzheimer’s research centers and may not fully reflect primary care populations. External validation on community-based cohorts is needed.
2. To assess population comparability with an external cohort, we compared baseline demographic and APOE epsilon4 distributions between NACC (n=37,633) and the Alzheimer’s Disease Neuroimaging Initiative (ADNI; n=2,049). Of our 101 predictors, only sex, race, ethnicity, age, and APOE epsilon4 carrier count had directly comparable fields in the public ADNIMERGE table; the comorbidity, functional, and behavioral variables that make up most of our feature set are not captured in ADNIMERGE, which is built around cognitive testing and imaging/biomarker measures rather than the detailed medical-history

assessment used in NACC’s UDS. Even restricted to these comparable fields, the cohorts differ appreciably: ADNI was less racially diverse (92.7% White vs. 80.9% in NACC) and had a lower proportion of Hispanic/Latino participants (3.2% vs. 8.1%), consistent with ADNI’s research/clinical-trial recruitment relative to NACC’s clinic-based ADRC population. Additionally, ADNI’s diagnostic categorization does not map directly onto the CDR-Global-based staging used here, precluding outcome-level comparison even where demographic fields align. Given this limited feature overlap, we did not attempt a reduced-feature model comparison against ADNI, as it would not constitute a meaningful external validation of the full model; genuine external validation will require a cohort with comparably detailed clinical and comorbidity data [9, 10].

3. **Minority Representation:** Sample sizes for some racial groups (American Indian/Alaska Native, Native Hawaiian/Pacific Islander) are limited, necessitating cautious interpretation of fairness metrics for these populations.
4. **Temporal Prediction Horizon:** We predict next-visit diagnosis, but the time interval between visits in the NACC cohort is variable, ranging from months to years depending on participant scheduling and retention. This variability is not explicitly modeled in the current framework; the model treats all next-visit predictions equivalently regardless of the elapsed time between assessments. Future work will adopt survival analysis approaches—such as Cox-Time or DeepSurv—to explicitly model variable time horizons and predict time-to-conversion rather than next-visit state alone.
5. **Feature Availability and Oversampling:** APOE4 genotyping may not be routinely available, and BorderlineSMOTE may generate biologically implausible combinations; sensitivity analysis under feature missingness and cost-sensitive alternatives warrant further study.

Future work will include: (1) external validation on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) cohort, (2) survival analysis for time-to-conversion prediction, (3) integration with electronic health record systems for prospective validation, and (4) development of a clinical decision support interface for end-user deployment.

429 5. Conclusion

430 We presented a robust and interpretable machine learning framework for
431 predicting dementia progression using exclusively routine clinical data. The
432 base XGBoost model achieved 87.82% accuracy, 85.14% macro F1-score, and
433 87.95% weighted F1-score on a large, multi-center cohort. Ablation analysis
434 showed that adding SMOTE or a more complex ensemble reduced aggregate
435 performance on this split, supporting a “Frugal AI” deployment strategy.

436 The framework’s key strengths include:

- 437 • **Clinical-Only Input:** No imaging, CSF, or genetic testing required
438 beyond routine assessments.
- 439 • **Rigorous Validation:** Patient-level data splitting prevents informa-
440 tion leakage and ensures honest performance estimates.
- 441 • **Fairness:** Performance was broadly similar across the evaluated sex,
442 age, and racial subgroups, though small subgroup sizes (e.g., N=13
443 Native Hawaiian/Pacific Islander, N=190 American Indian/Alaska Na-
444 tive) preclude definitive fairness conclusions for those groups.
- 445 • **Interpretability:** SHAP-based feature attribution enables clinician
446 understanding and regulatory compliance.
- 447 • **Error Severity:** Error severity analysis shows that most misclassifi-
448 cations remain clinically proximal (94.9% one stage off), although this
449 does not by itself establish clinical safety absent a prospective decision-
450 impact study.

451 This work supports a “Frugal AI” paradigm, democratizing access to early
452 Alzheimer’s detection in resource-constrained environments and bridging the
453 gap between theoretical AI advances and practical clinical tools.

454 CRedit Authorship Contribution Statement

455 **MD Walid Waccub Swadhin:** Conceptualization, Methodology, Soft-
456 ware, Formal analysis, Data Curation, Writing – Original Draft, Visualiza-
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490 Data Availability Statement

491 The NACC dataset is available through application to the National Alzheimer’s
492 Coordinating Center (<https://naccdata.org/>). Data used in this study re-
493 flect the September 2025 NACC data freeze (Data Request #16386). The

complete source code, trained model pipeline, and reproducibility materials for this study are publicly available at <https://github.com/walidsw/nacc-dementia-progression-ai>

Conflicts of Interest

The authors declare no conflicts of interest.

Ethics Statement

This study used de-identified secondary data from the National Alzheimer’s Coordinating Center (NACC) Uniform Data Set. No new participant recruitment, direct participant contact, intervention, or identifiable-data collection was undertaken by the authors; therefore, additional institutional review board approval was not required for this analysis. The original contributing Alzheimer’s Disease Research Centers obtained institutional review board approval and informed consent under their respective protocols. The analysis was conducted under the NACC Data Use Agreement associated with Data Request #16386 (September 2025 UDS data freeze).

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Declaration of Generative AI and AI-assisted Technologies

During manuscript preparation, the authors used GitHub Copilot and ChatGPT for language organization, drafting assistance, and code/document organization. The authors reviewed and edited all AI-assisted content, verified the reported results against the analysis outputs, and take full responsibility for the final manuscript.

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